

R2 Newton's second law

Name: _____

Class: _____

Date: _____

Time: **414 minutes**

Marks: **339 marks**

Comments:

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Q1.

A particle travels along a straight line. At time t , the velocity of the particle is v . When $t = 0$,

$v = u$. The acceleration of the particle throughout the motion is $\frac{k}{v}$, where k is a constant.

Show that

$$v^2 = u^2 + 2kt$$

(Total 6 marks)

Q2.

A particle P , of mass 0.25 kg, moves in a horizontal plane so that at time t seconds its velocity $\mathbf{v}$ m s⁻¹ is given by

$$\mathbf{v} = 8 \sin t \mathbf{i} + 4 \cos t \mathbf{j}$$

- (a) Find the force, $\mathbf{F}$ newtons, acting on P at time t .

(3)

- (b) (i) Show that the magnitude of $\mathbf{F}$ at time t is

$$\sqrt{(3 \cos^2 t + 1)}$$

(3)

- (ii) Hence state the range of values of the magnitude of $\mathbf{F}$.

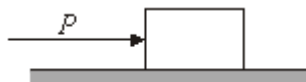
(2)

(Total 8 marks)

Q3.

A small block, of mass 2 kg, is on a rough horizontal surface. The coefficient of friction

between the block and the surface is $\frac{1}{7}$. A horizontal force of magnitude P newtons is applied to the block, as shown in the diagram.



- (a) The magnitude of the frictional force that acts between the block and the surface is F newtons. Show that the maximum possible value of F is 2.8.

(2)

- (b) The value of P increases gradually from 0 to 5.

- (i) When $P = 1$, state the value of F .

(1)

- (ii) When $P = 5$, the block moves with constant acceleration $a \text{ m s}^{-2}$. Find the value of a .

(4)

(Total 7 marks)

Q4.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

- (a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.

- (i) Draw a diagram showing the forces acting on the car.

(2)

- (ii) Show that $R = 660$.

(4)

- (iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

(1)

- (b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

- (i) Find the magnitude of the retardation of the car in this stage.

(2)

- (ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

(2)

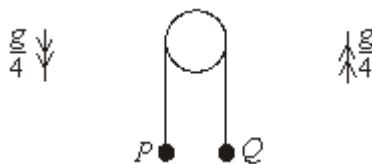
(Total 11 marks)

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Q5.

Two particles, P and Q , are attached to the ends of a light inextensible string which hangs over a smooth fixed peg, as shown in the diagram. The masses of P and Q are $5m$ and km respectively, where $k < 5$.

The particles are released from rest and move with acceleration of magnitude $\frac{g}{4}$.



- (a) Find, in terms of m and g , the tension in the string.

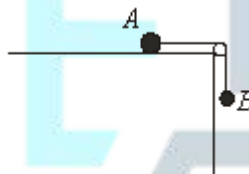
- (3)
- (b) Find the value of k . (3)
- (c) Find the magnitude of the force on the peg due to the string. (1)
- (d) The particles are released from rest at the same horizontal level when $t = 0$.
 Find, in terms of g , the vertical distance between the particles when $t = \frac{2}{3}$. (3)
- (Total 10 marks)

Q6.

A particle A of mass 0.2 kg is connected to a particle B of mass 0.3 kg by a light inextensible string.

The particle A is placed on a rough horizontal surface.

The string passes over a smooth fixed peg with B hanging freely, as shown in the diagram.



- (a) The system is held in equilibrium by a horizontal force, P newtons, which is applied to the particle A . The coefficient of friction between A and the rough horizontal surface is 0.5 .
- (i) Draw a diagram showing the forces acting on the particle A . (2)
- (ii) Show that the magnitude of the maximum possible frictional force between A and the rough horizontal surface is 0.98 newtons. (2)
- (iii) Find the tension in the string. (2)
- (iv) Find the least value of P required to hold the system in equilibrium. (2)
- (b) The horizontal force P is removed and the particles start to move from rest.
- (i) Find the magnitude of the acceleration of the particles during the subsequent motion.

(5)

- (ii) After falling 0.1 metres, the particle B hits the ground. Find the time between the system being released and B hitting the ground.

(2)

(Total 15 marks)

Q7.

A particle P , of mass 0.5 kg, moves under the action of a force $\mathbf{F}$ newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

- (a) Find the acceleration of P at time t .

(1)

- (b) The particle P starts its motion from rest.

Find the velocity of P at time t .

(2)

- (c) The particle P starts its motion at the point O .

- (i) Find the position vector of P relative to O at time t .

(3)

- (ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)

Q8.

John drops a rock of mass m kg from the top of a vertical cliff. At time t seconds after the rock is dropped, it has speed v m s⁻¹ and the air resistance to its motion is of magnitude $0.1mv$ newtons.

- (a) Show that the differential equation connecting v and t for this motion is $\frac{dv}{dt} = g - 0.1v$.

(2)

- (b) Solve this differential equation to find v in terms of g and t .

(5)

- (c) Hence show that the speed of the rock cannot exceed 98 m s⁻¹.

(1)

(Total 8 marks)

Q9.

A child kicks a small toy brick in a straight line across a horizontal floor. The brick initially

moves at 3.5 m s^{-1} and comes to rest in a distance of 2.5 metres.

- (a) Show that the magnitude of the retardation of the brick is 2.45 m s^{-2} .

(3)

- (b) The mass of the brick is 0.2 kg.

- (i) Find the magnitude of the frictional force acting on the brick.

(2)

- (ii) Find the coefficient of friction between the brick and the floor.

(2)

(Total 7 marks)

Q10.

A mountain railway train moves on a straight track. The mass of the train and its passengers is 1000 kg. During its motion the train moves under the action of a variable propulsive force, P newtons, and a constant resistance force of R newtons.

- (a) During the first stage of its motion, the train moves horizontally with acceleration 0.25 m s^{-2} . In this stage, the value of P is 1200.

Show that $R = 950$.

(3)

- (b) During the second stage of its motion, the train moves up a slope inclined at an angle α to the horizontal, where $\sin \alpha = 0.1$.



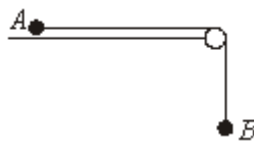
In this stage, the value of $P = 2100$, and the value of R remains at 950. Find the acceleration of the train.

(4)

(Total 7 marks)

Q11.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle A , of mass 0.6 kg, is in contact with a smooth horizontal surface, and the

particle B , of mass 0.1 kg , hangs freely above the ground. The system is released from rest with the string taut and A moves towards the peg.

It can be assumed that, during the subsequent motion, A does **not** reach the peg.

- (a) While the particles move freely, the string is taut.
- (i) Show that the acceleration of the particles is 1.4 m s^{-2} . (5)
- (ii) Find the tension in the string. (1)
- (iii) Find the magnitude of the resultant force on the peg due to the tension in the string. (3)
- (b) After q seconds of the motion, the particle B hits the ground and remains there. The string connecting A and B slackens and A continues to move towards the peg. Sketch a velocity–time graph to show the two stages of the motion of A after being released from rest. (3)

(Total 12 marks)

Q12.

Two particles, A and B , are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle A is of mass 0.5 kg and the particle B is of mass 0.1 kg . The particle A is in contact with a rough horizontal surface.

- (a) The system is at rest and the particle B hangs vertically.
- (i) Show that the tension in the string is 0.98 newtons . (1)
- (ii) The particle A rests in limiting equilibrium. Show that the coefficient of friction between A and the surface is 0.2 . (3)
- (iii) Draw a diagram to show the forces acting on the peg due to the string. (1)

(iv) Find the magnitude of the resultant force on the peg due to the string.

(2)

(b) An additional mass of 0.1 kg is attached to the particle B and the system is released from rest. The particle A moves across the surface towards the peg and B moves vertically downwards. The particles move with acceleration $a \text{ m s}^{-2}$ and the tension in the string is T newtons.

(i) Show that $a = 1.4$.

(5)

(ii) Find the value of T .

(1)

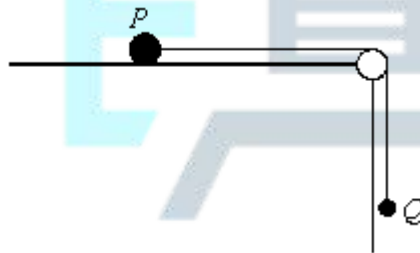
(iii) After falling a distance of 0.7 metres, B hits the ground. Find the time between the particles being released and B hitting the ground.

(2)

(Total 15 marks)

Q13.

Two particles, P and Q , are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle P is of mass $2m$ and Q is of mass m . The particle P is in contact with a horizontal surface, which may be modelled as either rough or smooth.

(a) In the first case, the surface is modelled as rough and the particle Q hangs at rest. The coefficient of friction between P and the surface is μ .

(i) Find the tension in the string.

(1)

(ii) Find the range of possible values of μ .

(3)

(b) In the second case, the surface is modelled as smooth. The system is released from rest with the string taut and the particle P moves towards the peg.

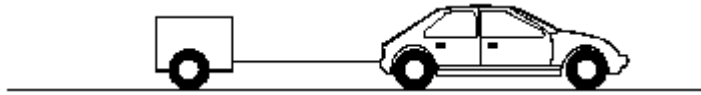
Find the tension in the string.

(5)

(Total 9 marks)

Q14.

The diagram shows a car pulling a trailer in a straight line on a horizontal stretch of road.



The mass of the car is 1250 kg and the total resistance force acting on the car is 500 N.

The mass of the trailer is 250 kg and the total resistance force acting on the trailer is 100 N.

- (a) During part of the journey, a constant braking force is applied to the car, causing the car and trailer to decelerate at a constant rate of 0.5 m s^{-2} .

- (i) By considering the forces on the trailer, find the magnitude of the force in the towbar between the car and the trailer.

(3)

- (ii) Find the magnitude of the braking force applied to the car.

(4)

- (b) Later in the journey, the car and trailer travel with constant speed.

State the magnitude of the tension in the towbar.

(1)

(Total 8 marks)

Q15.

A small case is moving on a rough horizontal surface. The case is being pushed with a force $(6\mathbf{i} - 2\mathbf{j})$ newtons, as shown in the diagram.



The case has mass 1.5 kg and moves with acceleration 0.6 m s^{-2} .

- (a) Draw a diagram to show the forces acting on the case.

(1)

- (b) Show that the frictional force acting on the case is of magnitude 5.1 newtons.

(3)

- (c) Find the normal reaction force between the case and the surface.

(3)

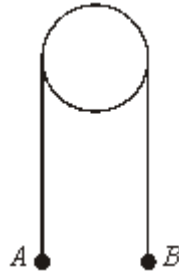
- (d) The coefficient of friction between the case and the surface is μ . Find the value of μ .

(2)

(Total 9 marks)

Q16.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth fixed peg, as shown in the diagram.



The particles A and B have masses 0.21 kg and 0.14 kg respectively. The system is released from rest. It may be assumed that, during the motion, neither particle reaches the peg.

- (a) Show that the magnitude of the acceleration of the particles during the subsequent motion is 1.96 m s^{-2} . (5)
 - (b) Find the speed of the particles after 2 seconds of the motion. (2)
 - (c) After 2 seconds of the motion, the string breaks and the particles move independently under the influence of gravity alone. Find the time between the string breaking and the particle B coming to instantaneous rest. (2)
 - (d) Sketch a velocity-time graph to show the motion of B between being released from rest and subsequently coming instantaneously to rest. (3)
- (Total 12 marks)

Q17.

A van accelerates uniformly from rest on a straight horizontal road. After it has travelled 80 metres, its speed is 20 m s^{-1} .

- (a)
 - (i) Find the time taken for the van to travel the 80 metres. (2)
 - (ii) Show that the acceleration of the van is 2.5 m s^{-2} . (2)
- (b) The mass of the van is 1200 kg . The force produced by its engine has magnitude F newtons and acts on the van in its direction of motion.
 - (i) Find F , if there is no resistance to motion.

(2)

- (ii) Find F , if there is a constant resistance force of 400 N.

(2)

(Total 8 marks)

Q18.

A child travels down a slide at a **constant speed**. Model the slide as a rough plane inclined at an angle of 40° to the horizontal. Model the child as a particle of mass 20 kg and assume that there is no air resistance as the child moves.

- (a) (i) Draw a diagram to show the forces acting on the child.

(1)

- (ii) Show that the magnitude of the normal reaction force acting on the child is approximately 150 N.

(2)

- (iii) Find the magnitude of the friction force acting on the child and show that the coefficient of friction between the child and the slide is 0.84, correct to two significant figures.

(4)

- (b) In reality, air resistance acts on the child as she slides at a constant speed. How would this affect the value of the coefficient of friction that you calculated in part (a)(iii)?

(1)

(Total 8 marks)

Q19.

A particle moves with constant acceleration in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the plane. At time $t = 0$, the particle is at rest at the origin. At time $t = 4$ seconds, the position vector of the particle is $(-16\mathbf{i} + 16\mathbf{j})$ metres.

- (a) Show that the acceleration of the particle is $(-2\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$.

(4)

- (b) Find the speed of the particle when $t = 5$.

(4)

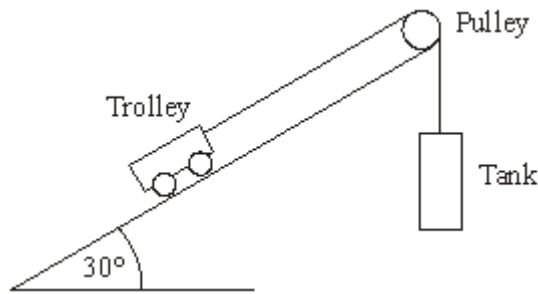
- (c) The mass of the particle is 3 kg. Two horizontal forces $\mathbf{P}$ and $\mathbf{Q}$ act on the particle. If $\mathbf{P} = 20\mathbf{i} - 10\mathbf{j}$, find $\mathbf{Q}$.

(4)

(Total 12 marks)

Q20.

A trolley, of mass 200 kg, is on a slope inclined at 30° to the horizontal. It is attached to a tank by a light, inextensible rope that passes over a smooth, light pulley. The tank can be filled with water. The situation is shown in the diagram below.



Model the tank and trolley as particles and assume that there is no resistance to motion.

- (a) When the tank is filled with water, the total mass of the tank and the water is 150 kg.

Show that the acceleration of the tank and trolley is 1.4 m s^{-2} .

(7)

- (b) When the tank is empty, the trolley can be set in motion, so that it moves with a constant speed up the slope. Find the mass of the empty tank.

(4)

(Total 11 marks)

Q21.

A car is travelling along a straight horizontal road. When the car is at a distance of 48 metres from a set of traffic lights, it is travelling at 20 m s^{-1} . The car slows down uniformly so that it has a speed of 4 m s^{-1} as it passes the traffic lights.

- (a) Show that the acceleration of the car is -4 m s^{-2} .

(2)

- (b) How far past the traffic lights does the car travel before it stops?

(3)

- (c) Calculate the total time that it takes the car to stop.

(2)

- (d) The car has mass 1100 kg. As the car slows down, it experiences a resistance force that has a constant magnitude of 200 newtons, and a force due to the brakes of magnitude F newtons.

Find F .

(3)

(Total 10 marks)

Q22.

A particle, of mass 2 kg, is set into motion up a rough slope inclined at 40° to the horizontal.

The coefficient of friction between the particle and the slope is 0.3. Assume that there is no air resistance acting on the particle.

- (a) Find the magnitude of the friction force that acts on the particle.

(3)

- (b) Find the acceleration of the particle as it moves up the slope.

(3)

(Total 6 marks)

Q23.

A lift has mass 800 kg. A vertical cable is attached to the lift. Model the lift as a particle.

- (a) Show that the tension in the cable is 8000 N, when the lift is accelerating upwards at 0.2 m s^{-2} .

(3)

- (b) Find the tension in the cable when the lift is accelerating downwards at 0.2 m s^{-2} .

(2)

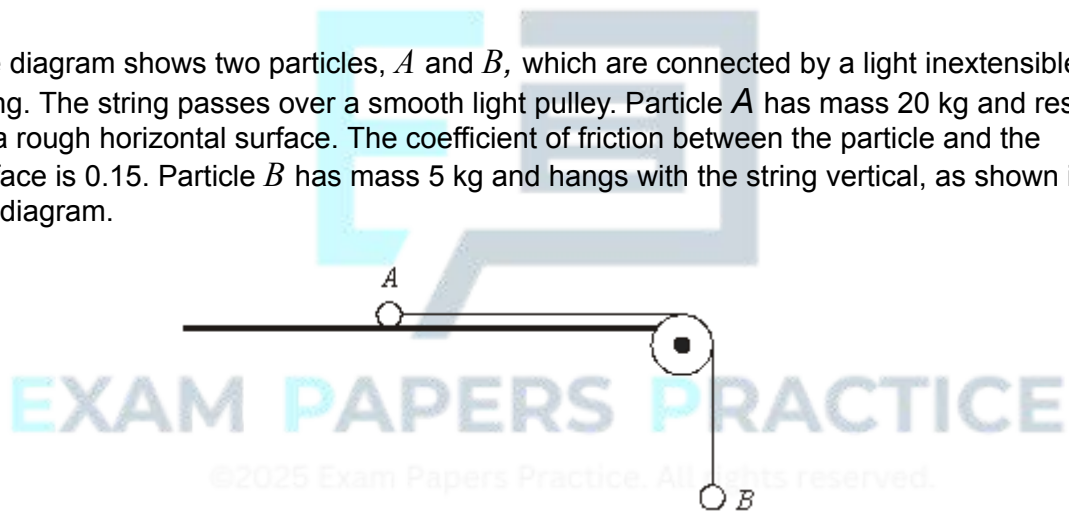
- (c) Write down the tension in the cable when the lift is travelling at a constant speed.

(1)

(Total 6 marks)

Q24.

The diagram shows two particles, A and B , which are connected by a light inextensible string. The string passes over a smooth light pulley. Particle A has mass 20 kg and rests on a rough horizontal surface. The coefficient of friction between the particle and the surface is 0.15. Particle B has mass 5 kg and hangs with the string vertical, as shown in the diagram.



The particles are released from rest and begin to move.

- (a) Find the magnitude of the friction force acting on A .

(2)

- (b) By forming an equation of motion for each particle, show that the acceleration of each particle has magnitude 0.784 m s^{-2} .

(5)

- (c) Find the tension in the string.

(2)

- (d) Find the distance that A moves in the first 3 seconds of its motion. Assume that the acceleration of A remains constant during this time.

(3)

(Total 12 marks)

Q25.

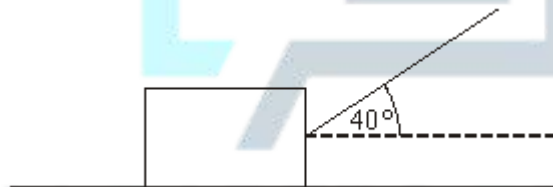
A cyclist travels along a straight horizontal road. She accelerates uniformly from a speed of 2 m s^{-1} to 6 m s^{-1} as she travels 10 metres.

- (a) (i) Show that the acceleration of the cyclist is 1.6 m s^{-2} . (2)
- (ii) Find the time that it takes the cyclist to travel this distance. (2)
- (b) Model the cyclist as a particle of mass 65 kg. A constant resistance force of magnitude 35 N acts on the cyclist. A horizontal force of magnitude F newtons also acts on the cyclist in the direction of motion. Find F . (3)

(Total 7 marks)

Q26.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is T newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.



Model the box as a particle.

- (a) Draw a diagram to show the forces acting on the box. (1)
- (b) Find, in terms of T , the magnitude of the normal reaction force acting on the box. (3)
- (c) Show that the magnitude of the friction force acting on the box is $294 - 0.6T \sin 40^\circ$. (2)
- (d) The box accelerates at 0.5 m s^{-2} . Find T . (4)

(Total 10 marks)

Q27.

A car travels along a straight horizontal road. When it passes a set of traffic lights it is travelling at a speed of 3 m s^{-1} . The car then accelerates at a constant 1.2 m s^{-2} until it reaches a speed of 9 m s^{-1} .

- (a) Show that the car accelerates for 5 seconds. (2)
 - (b) Find the distance that the car travels as it accelerates from 3 m s^{-1} to 9 m s^{-1} . (2)
 - (c) The mass of the car is 1200 kg. Calculate the magnitude of the resultant force acting on the car while it is accelerating. (2)
- (Total 6 marks)**

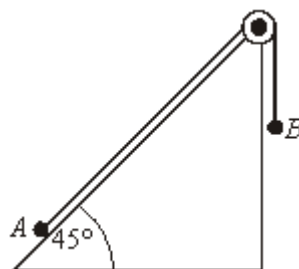
Q28.

A rough plane is inclined at an angle of 40° to the horizontal. A particle, of mass 5 kg, is sliding down the plane.

- (a) Draw a diagram to show the forces acting on the particle. (1)
 - (b) Find the magnitude of the normal reaction force acting on the particle. (2)
 - (c) The coefficient of friction between the particle and the plane is 0.2. Show that the magnitude of the friction force acting on the particle is 7.51 N, correct to three significant figures. (2)
 - (d) Show that the acceleration of the particle is 4.80 m s^{-2} , correct to three significant figures. (4)
 - (e) Find the distance that the particle travels as its speed increases from 2 m s^{-1} to 10 m s^{-1} . (3)
- (Total 12 marks)**

Q29.

Two particles, *A* and *B*, are connected by a light inextensible string, which passes over a smooth light pulley. Particle *A* is on a smooth slope, at 45° to the horizontal, and particle *B* hangs with the string vertical, as shown in the diagram.



The mass of A is 14 kg and the mass of B is 6 kg.

- (a) Using two equations of motion, show that the acceleration of the particles is 1.91 m s^{-2} correct to three significant figures. (6)
- (b) Particle B is replaced by a particle C of mass m kg. After the particles have been set in motion, they move with a constant speed. Find m . (4)

(Total 10 marks)

Q30.

A crate, of mass 50 kg, is at rest on a warehouse floor. The floor is rough and horizontal. The coefficient of friction between the crate and the floor is μ . A rope is attached to the crate at an angle of 30° to the horizontal. The tension in the rope is 100 N. The crate is shown in the diagram.



Model the crate as a particle.

- (a) Draw and label a diagram to show the forces acting on the crate. (1)
- (b) Show that the magnitude of the normal reaction force acting on the crate is 440 N. (3)
- (c) If the crate remains at rest, μ must satisfy the inequality $\mu \geq k$. Find k . (3)
- (d) If $\mu = 0.1$, find the acceleration of the crate. (4)

(Total 11 marks)

Q31.

As a lift moves upwards its motion has three stages.

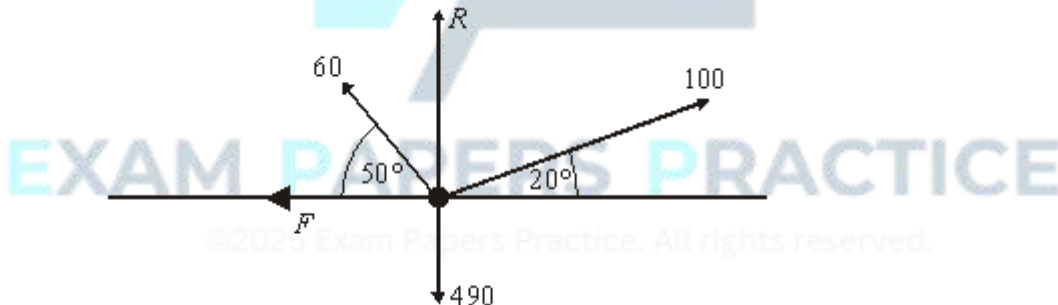
Stage I It accelerates uniformly, from rest, at 0.2 m s^{-2} , for 8 seconds.
Stage II It then travels at a constant speed for 3 seconds.
Stage III It then decelerates uniformly, stopping when it has travelled a further 1 metre.

- (a) (i) Find the distance travelled by the lift in stage I. (2)
- (ii) Find the speed of the lift at the end of stage I. (2)
- (iii) Find the total distance travelled by the lift. (2)
- (b) Find the total time for which the lift is moving. (3)
- (c) The lift and its passengers have a total mass of 600 kg. The lift is supported by a single cable. Find the tension in the cable when the lift is accelerating upwards at 0.2 m s^{-2} . (3)

(Total 12 marks)

Q32.

A box, of mass 50 kg, moves in a straight line on a rough horizontal surface. The diagram below shows **all** the forces acting on the box, as it moves. The magnitude of each force is in newtons. The box is modelled as a particle.



Note that the weight of the box has been included in the diagram.

- (a) Find R , the magnitude of the normal reaction force acting on the box. (3)
- (b) The box accelerates at 0.5 m s^{-2} .
- (i) Show that F , the magnitude of the friction force acting on the box, is approximately 30.4. (4)
- (ii) Find the coefficient of friction between the box and the surface. (2)

(Total 9 marks)

Q33.

A particle moves in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Initially the particle is at the origin and has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration. At time $t = 10$ it has position vector $(40\mathbf{i} - 15\mathbf{j})$ metres.

- (a) Show that the velocity of the particle is $(3\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$ when $t = 10$. (4)
- (b) Find the acceleration of the particle. (3)
- (c) The mass of the particle is 15 kg. Find the magnitude of the resultant force acting on the particle. (4)
- (Total 11 marks)**

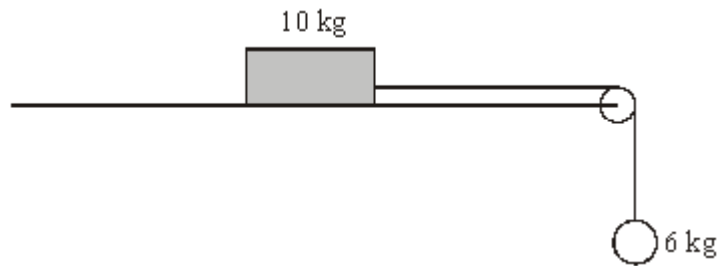
Q34.

A box, of mass 20 kg, is initially at rest on a rough horizontal surface. A horizontal force of magnitude P newtons is applied to the box. The coefficient of friction between the box and the surface is 0.3.

- (a) State the magnitude of the normal reaction force acting on the box. (1)
- (b) Find the magnitude of the friction force that acts on the box if:
- (i) $P = 80$;
- (ii) $P = 40$. (3)
- (c) Find the value of P when the box is accelerating at 0.8 m s^{-2} . (3)
- (d) When the box reaches a speed of 6 m s^{-1} , the horizontal force P is removed. Find the distance that the box travels after the force P is removed. (5)
- (Total 12 marks)**

Q35.

A block, of mass 10 kg, rests on a rough horizontal surface. It is connected by a light, inextensible string to a particle of mass 6 kg. The string passes over a light, smooth pulley, so that the string hangs vertically, as shown in the diagram.



Model the block as a particle.

- (a) The system is released from rest and the block travels 0.5 metres in 2 seconds. Find the acceleration of the system.

(2)

- (b) Show that the tension in the string is 57.3 N.

(3)

- (c) Find the coefficient of friction between the block and the plane.

(6)

(Total 11 marks)



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